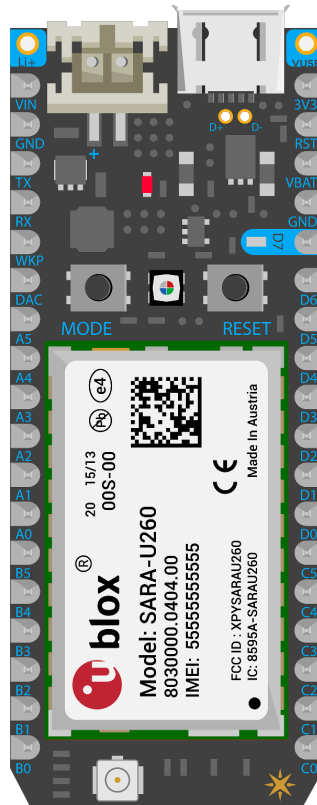


Electron Datasheet

The Electron has been deprecated. See the [Supply Secure FAQ](#) for more information.



Functional description

OVERVIEW

Note: The Electron U260 and U270 have been deprecated and are no longer available for sale. New designs should use the B-Series SoM or Boron. For a pin-compatible replacement for the G350, U260, and U270, the Electron ELC314 is available in tray quantities only.

The Electron is a tiny development kit for creating cellular-connected electronics projects and products. It comes with a SIM card (Nano 4FF)^[1] and an affordable data plan for low-bandwidth things. Plus it's available for more than 100 countries worldwide!

It also comes with Particle's development tools and cloud platform for managing and interacting with your new connected hardware.

FEATURES

- U-blox SARA-U201/U260/U270 (3G with 2G fallback), G350 (2G), or R410M (LTE Cat M1) cellular module
- STM32F205RGT6 120MHz ARM Cortex M3 microcontroller
- 1MB flash, 128KB RAM
- BQ24195 power management unit and battery charger
- MAX17043 fuel gauge
- RGB status LED
- 30 mixed-signal GPIO and advanced peripherals
- Open source design
- Real-time operation system (RTOS)
- FCC (United States), CE (European Union), and ISED (Canada) certified

^[1]The LTE model uses a MFF2 SMD Particle SIM instead of a physical SIM card. The Electron LTE is only available to existing enterprise customers who have deployed an Electron 2G/3G solution and would like to upgrade to LTE. It is only available in tray quantities. New designs should use the Boron LTE or B-Series B404 SoM. It can only be used in the United States, Canada, and Mexico at this time.

DEVICE OS SUPPORT

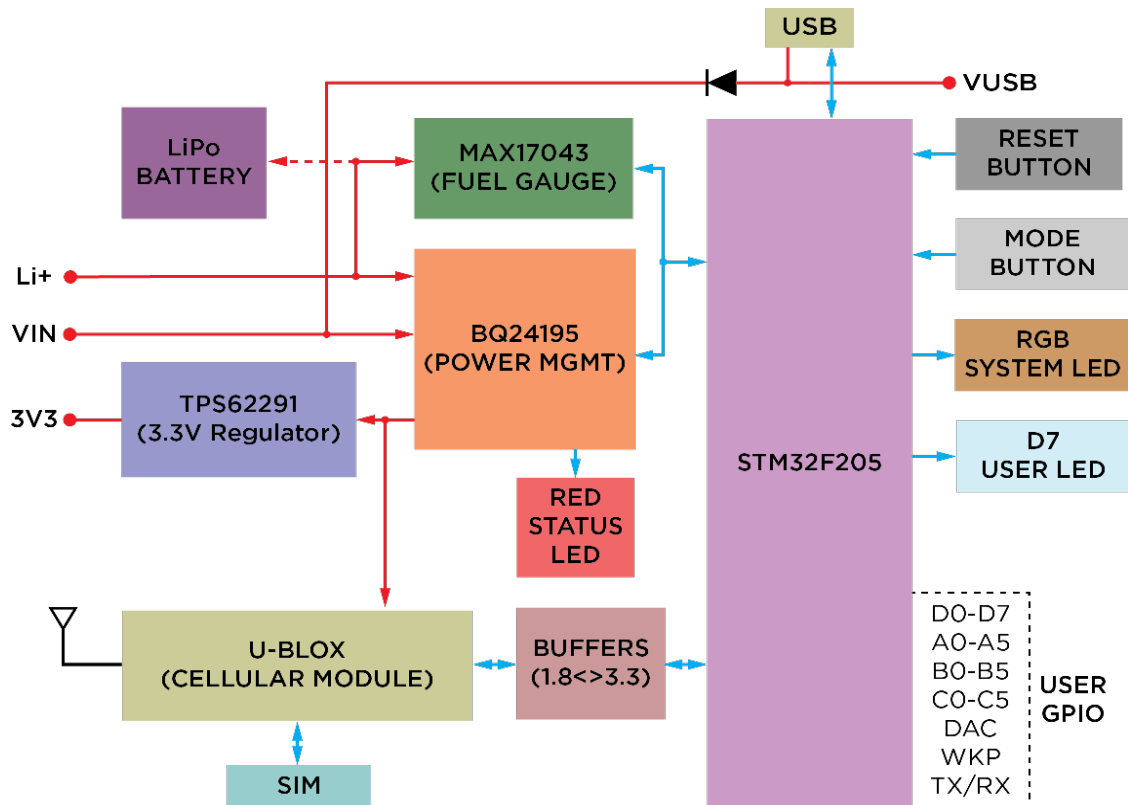
It is recommended that you use the latest version in the 2.x LTS release line with the all Electrons.

While the devices are compatible with older versions of Device OS and 3.x, these versions as past the end-of-support date and are not recommended for use in production. Only 2.x LTS remains in the Extended Support and Maintenance (ESM) window.

For information on upgrading Device OS, see [Version information](#). For the latest version shipped from the factory, see [Manufacturing firmware versions](#) page. See also [Long Term Support \(LTS\) releases](#).

Interfaces

BLOCK DIAGRAM



POWER

The Electron can be powered via the VIN (3.9V-12VDC) pin, the USB Micro B connector or a LiPo battery.

USB

Most USB ports can supply only a maximum of 500mA, but the u-Blox GSM module on the Electron alone can consume a peak of 800mA to 1800mA (2G/3G) or 490 mA (LTE) of current during transmission. In order to compensate of this deficit, one must connect the LiPo battery at all times when powering from a traditional USB port for 2G/3G. The Electron will intelligently source power from the USB most of the time and keep the battery charged. During peak current requirements, the additional power will be sourced from the battery. This reduces the charge-discharge cycle load on the battery, thus improving its longevity.

VIN

The input voltage range on VIN pin is 3.9VDC to 12VDC. When powering from the VIN pin alone, make sure that the power supply is rated at 10W (for example 5VDC at 2Amp). If the power source is unable to meet this requirement, you'll need connect the LiPo battery as well. An additional bulk capacitance of 470uF to 1000uF should be added to the VIN input when the LiPo Battery is disconnected. The amount of capacitance required will depend on the ability of the power supply to deliver peak currents to the cellular modem.

The Electron LTE (ELC404 & ELC402, LTE Cat 1) can be powered with as little as 550 mA at 5V.

LiPo Battery

When powered from a LiPo battery alone, the power management IC switches off the internal

regulator and supplies power to the system directly from the battery. This reduces the conduction losses and maximizes battery run time. The battery provided with the Electron is a Lithium-Ion Polymer battery rated at 3.7VDC 1,800mAh. You can substitute this battery with another 3.7V LiPo with higher current rating. Remember to never exceed this voltage rating and always pay attention to the polarity of the connector.

Typical current consumption is around 180mA and up to 1.8A transients at 5VDC. In deep sleep mode, the quiescent current is 130uA (powered from the battery alone).

Li+

This pin is internally tied to the positive terminal of the LiPo battery connector. It is intentionally left unpopulated. Please note that an incorrect usage of this pin can render the Electron unusable.

Li+ pin serves two purposes. You can use this pin to connect a LiPo battery directly without having to use a JST connector or it can be used to connect an external DC power source (and this is where one needs to take extra precautions). When powering it from an external regulated DC source, the recommended input voltage range on this pin is between 3.6V to 4.4VDC. Make sure that the supply can handle currents of at least 2 Amps.

This is the most efficient way of powering the Electron since the PMIC by-passes the regulator and supplies power to the Electron via an internal FET leading to lower quiescent current.

VUSB

This pin is internally connected to USB supply rail and will output 5V when the Electron is plugged into an USB port. It is intentionally left unpopulated. This pin will *NOT* output any voltage when the Electron is powered via VIN and/or the LiPo battery.

3V3 Pin

This pin is the output of the on-board 3.3V switching regulator that powers the microcontroller and the peripherals. This pin can be used as a 3.3V power source with a max load of 800mA. 3.3V will also be available on that pin while the device being in deep sleep. Unlike the Photon or the Core, this pin *CANNOT* be used as an input to power the Electron.

VBAT

Supply to the internal RTC, backup registers and SRAM when 3V3 is not present (1.65 to 3.6VDC). The Pin is internally connected to 3V3 supply via a 0 ohm resistor. If you wish to power VBAT via an external supply, you'll need to remove this resistor with a desoldering iron. Contact us if you wish to request trays of electrons with this jumper depopulated.

Powering the Electron without a battery

The most forgiving way to power the Electron without a battery is via the VIN input [see VIN above](#). Power may also be applied separately to the Li+ pin or LiPo JST connector [see Li+ above](#).

FCC APPROVED ANTENNAS

Antenna Type	Manufacturer	MFG. Part #	Gain
2G/3G PCB antenna	Taoglas	PC104.07.0165C	1dBi ~ 2.39dBi
LTE flex antenna	Taoglas	FXUB63.07.0150C	5.00dBi peak

Peripheral Type	Qty	Input(I) / Output(O)	FT ^[1] / 3V3 ^[2]
Digital	30	I/O	FT/3V3
Analog (ADC)	12	I	3V3
Analog (DAC)	2	O	3V3
UART	3	I/O	3V3
SPI	2	I/O	3V3
I2S	1	I/O	3V3
I2C	1	I/O	FT
CAN	2	I/O	3V3 ^[4]
USB	1	I/O	3V3
PWM	13 ³	O	3V3

Notes: ^[1] FT = 5.0V tolerant pins. All pins except A3 and DAC are 5V tolerant (when not in analog mode). If used as a 5V input the pull-up/pull-down resistor must be disabled.

^[2] 3V3 = 3.3V max pins.

^[3] PWM is available on D0, D1, D2, D3, B0, B1, B2, B3, A4, A5, WKP, RX, TX with a caveat: PWM timer peripheral is duplicated on two pins (A5/D2) and (A4/D3) for 11 total independent PWM outputs. For example: PWM may be used on A5 while D2 is used as a GPIO, or D2 as a PWM while A5 is used as an analog input. However A5 and D2 cannot be used as independently controlled PWM outputs at the same time.

^[4] Technically these pins are 5.0V tolerant, but since you wouldn't operate them with a 5.0V transceiver it's proper to classify them as 3.3V.

JTAG AND SWD

Pin D3 through D7 are JTAG interface pins. These can be used to reprogram your Electron bootloader or user firmware image with standard JTAG tools such as the ST-Link v2, J-Link, R-Link, OLIMEX ARM-USB-TINY-H, and also the FTDI-based Particle JTAG Programmer. If you are short on available pins, you may also use SWD mode which requires less connections.

| Electron Pin | JTAG | SWD | STM32F205RGT6 Pin | Default Internal^[1] | | : : | : : | : : | : : | D7 | JTAG_TMS | SWD/SWDIO | PA13 | ~40k pull-up | | D6 | JTAG_TCK | CLK/SWCLK | PA14 | ~40k pull-down | | D5 | JTAG_TDI | | PA15 | ~40k pull-up | | D4 | JTAG_TDO | | PB3 | Floating | | D3 | JTAG_TRST | | PB4 | ~40k pull-up | | 3V3 | Power | | | | GND | Ground | | | | RST | Reset | | | |

Notes: ^[1] Default state after reset for a short period of time before these pins are restored to GPIO (if JTAG debugging is not required, i.e. `USE_SWD_JTAG=y` is not specified on the command line.)

Memory map

STM32F205RGT6 FLASH LAYOUT OVERVIEW

- Bootloader (16 KB)
- DCD1 (16 KB), stores keys, mfg info, system flags, etc..
- DCD2 (16 KB), swap area for DCD1
- EEPROM emulation bank 1 (16 KB)
- EEPROM emulation bank 2 (64 KB)
- Device OS (512 KB) [256 KB comms + 256 KB hal/platform/services]
- Factory backup, OTA backup and user application (384 KB) [3 x 128 KB]

DCD LAYOUT

The DCD area of flash memory has been mapped to a separate DFU media device so that we can incrementally update the application data. This allows one item (say, server public key) to be updated without erasing the other items.

DCD layout in *release/stable* [found here in firmware.](#)

Region	Offset	Size
system flags	0	32
version	32	2
device private key	34	1216
device public key	1250	384
ip config	1634	120
feature flags	1754	4
country code	1758	4
claim code	1762	63
claimed	1825	1
ssid prefix	1826	26
device code	1852	6
version string	1858	32
dns resolve	1890	128
reserved1	2018	64
server public key	2082	768
padding	2850	2
flash modules	2852	100
product store	2952	24
antenna selection	2976	1
cloud transport	2977	1
alt device public key	2978	128
alt device private key	3106	192
alt server public key	3298	192
alt server address	3490	128
device id	3618	12
radio flags	3630	1
mode button mirror	3631	32

led mirror	3663	96
led theme	3759	64
reserved2	3823	435

MEMORY MAP (COMMON)

Region	Start Address	End Address	Size
Bootloader	0x8000000	0x8004000	16 KB
DCD1	0x8004000	0x8008000	16 KB
DCD2	0x8008000	0x800C000	16 KB
EEPROM1	0x800C000	0x8010000	16 KB
EEPROM2	0x8010000	0x8020000	64 KB

MEMORY MAP (MODULAR FIRMWARE - DEFAULT)

Before 0.6.0 firmware

Region	Start Address	End Address	Size
System Part 1	0x8020000	0x8040000	128 KB
System Part 2	0x8040000	0x8060000	128 KB
Application	0x8080000	0x80A0000	128 KB
Factory Reset/Extended Application	0x80A0000	0x80C0000	128 KB
OTA Backup	0x80C0000	0x80E0000	128 KB
Decompress region	0x80E0000	0x8100000	128 KB

Since 0.6.0 firmware

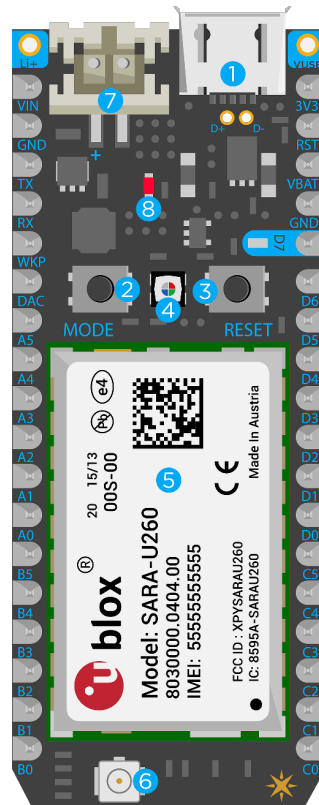
Region	Start Address	End Address	Size
System Part 2	0x8020000	0x8040000	128 KB
System Part 3	0x8040000	0x8060000	128 KB
System Part 1	0x8060000	0x8080000	128 KB
Application	0x8080000	0x80A0000	128 KB
Factory Reset/Extended Application	0x80A0000	0x80C0000	128 KB
OTA Backup	0x80C0000	0x80E0000	128 KB
Decompress region	0x80E0000	0x8100000	128 KB

MEMORY MAP (MONOLITHIC FIRMWARE - OPTIONAL)

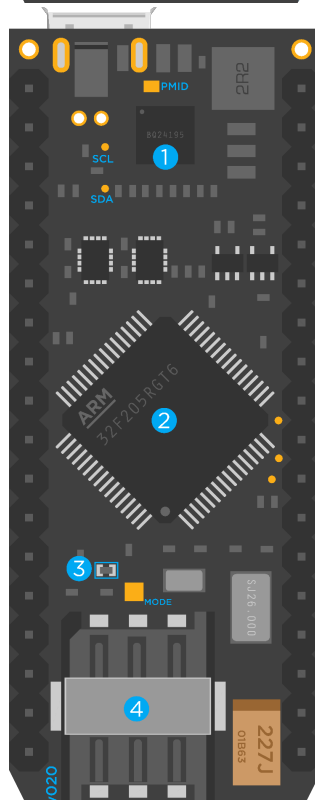
Region	Start Address	End Address	Size
Firmware	0x8020000	0x8080000	384 KB
Factory Reset	0x8080000	0x80E0000	384 KB
Unused (factory reset modular)	0x80E0000	0x8100000	128 KB

Pin and button definition

PIN MARKINGS:



- 1 USB
- 2 SETUP BUTTON
- 3 RESET BUTTON
- 4 RGB LED
- 5 u-blox CELL MODULE
- 6 u-FL ANT CONN.
- 7 LiPo BATT CONN.
- 8 CHARGE STATUS LED



- 1 POWER MGMT IC
- 2 MICROCONTROLLER
- 3 3V3-VBAT JUMPER
- 4 SIM CARD HOLDER

PIN DESCRIPTION

Pin	Description
VIN	This pin can be used as an input or output. As an input, supply 5VDC to 12VDC to power the Electron. When the Electron is powered via the USB port, this pin will output a voltage of approximately 4.8VDC due to a reverse polarity protection series Schottky diode between VUSB and VIN. When used as an output, the max load on VIN is 1Amp.
RST	Active-low reset input. On-board circuitry contains a 10k ohm pull-up resistor between RST and 3V3, and 0.1uF capacitor between RST and GND.
VBAT	Supply to the internal RTC, backup registers and SRAM when 3V3 is not present (1.65 to 3.6VDC). The Pin is internally connected to 3V3 supply via a 0 ohm resistor. If you wish to power VBAT via an external supply, you'll need to remove this resistor with a desoldering iron. Contact us if you wish to request trays of electrons with this jumper depopulated.
3V3	This pin is the output of the on-board regulator. When powering the Electron via VIN or the USB port, this pin will output a voltage of 3.3VDC. The max load on 3V3 is 800mA. It should not be used as an input to power the Electron.
WKP	Active-high wakeup pin, wakes the module from sleep/standby modes. When not used as a WAKEUP, this pin can also be used as a digital GPIO, ADC input or PWM ^[1] . Can be referred to as A7 when used as an ADC.
DAC	12-bit Digital-to-Analog (D/A) output (0-4095), referred to as DAC or DAC1 in software. Can also be used as a digital GPIO or ADC. Can be referred to as A6 when used as an ADC.
RX	Primarily used as UART RX, but can also be used as a digital GPIO or PWM ^[1] .
TX	Primarily used as UART TX, but can also be used as a digital GPIO or PWM ^[1] .
D0-D7	Digital only GPIO. D0 , D1 , D2 , D3 can also be used as PWM ^[1] outputs.
A0-A7	12-bit Analog-to-Digital (A/D) inputs (0-4095), and also digital GPIOs. A6 and A7 are code convenience mappings, which means pins are not actually labeled as such but you may use code like <code>analogRead(A7)</code> . A6 maps to the DAC pin and A7 maps to the WKP pin. A3 is also a second DAC output used as DAC2 or A3 in software. A4 and A5 can also be used as PWM ^[1] outputs.
B0-B5	B0 and B1 are digital only while B2 , B3 , B4 , B5 are 12-bit A/D inputs as well as digital GPIOs. B0 , B1 , B2 , B3 can also be used as PWM ^[1] outputs.
C0-C5	Digital only GPIO. C4 and C5 can also be used as PWM ^[1] outputs.
VUSB	This pin is internally connected to USB supply and will output 5V when the Electron is plugged into an USB port. It is intentionally left unpopulated.
Li+	This pin is internally connected to the positive terminal of the LiPo battery. It is intentionally left unpopulated.

Notes:

^[1] PWM is available on D0, D1, D2, D3, B0, B1, B2, B3, A4, A5, WKP, RX, TX with a caveat: PWM timer peripheral is duplicated on two pins (A5/D2) and (A4/D3) for 11 total independent PWM outputs. For example: PWM may be used on A5 while D2 is used as a GPIO, or D2 as a PWM while A5 is used as an analog input. However A5 and D2 cannot be used as independently controlled PWM outputs at the same time.

LED STATUS

Charge status LED

State	Description
ON	Charging in progress
OFF	Charging complete
Blink at 1Hz	Fault condition ^[1]
Rapid blinking	Battery disconnected ^[2]

Notes:

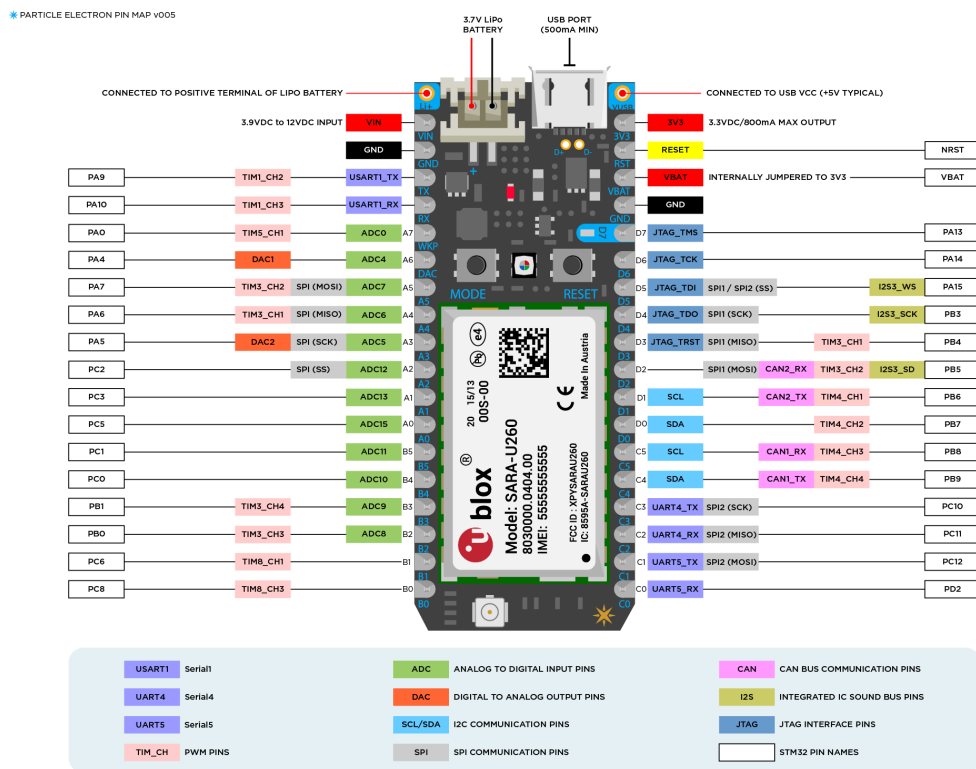
[1] A fault condition can occur due to several reasons, for example, battery over/under voltage, temperature fault or safety timer fault. You can find the root cause by reading the fault register of the power management IC in firmware.

[2] You can stop this behavior by either plugging in the LiPo battery or by disabling charging using firmware command: `PMIC().disableCharging();`.

System RGB LED

For a detailed explanation of different color codes of the RGB system LED, please take a look [here](#).

PINOUT DIAGRAM



You can download a high resolution pinout diagram in a [PDF version here](#).

Technical specifications

ABSOLUTE MAXIMUM RATINGS ^[1]

Parameter	Symbol	Min	Typ	Max	Unit
Supply Input Voltage	V _{IN-MAX}			+17	V
Supply Output Current	I _{IN-MAX-L}			1	A
Battery Input Voltage	V _{LiPo}			+6	V
Supply Output Current	I _{3V3-MAX-L}			800	mA
Storage Temperature	T _{stg}	-30		+75	°C
ESD Susceptibility HBM (Human Body Mode)	V _{ESD}			2	kV

^[1] Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under recommended operating conditions is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

RECOMMENDED OPERATING CONDITIONS

Parameter	Symbol	Min	Typ	Max	Unit
Supply Input Voltage	V _{VIN}	+3.88 ^[1]		+12	V
Supply Output Voltage	V _{VIN}		+4.8		V
Supply Output Voltage	V _{3V3}		+3.3		V
LiPo Battery Voltage	V _{LiPo}	+3.6		+4.4	V
Supply Input Voltage	V _{VBAT}	+1.65		+3.6	V
Supply Input Current (VBAT)	I _{VBAT}			19	uA
Operating Current (uC on, Cellular ON)	I _{VIN avg}		180	250	mA
2G/3G Peak Current (uC on, Cellular ON)	I _{IN pk}	800 ^[2]		1800 ^[3]	mA
LTE Peak Current (uC on, Cellular ON)	I _{IN pk}		550		mA
Operating Current (uC on, Cellular OFF)	I _{VIN avg}		47	50	mA
Sleep Current (4.2V LiPo, Cellular OFF)	I _{Qs}		0.8	2	mA
Deep Sleep Current (4.2V LiPo, Cellular OFF)	I _{Qds}		110	130	uA
Operating Temperature	T _{op}	-20		+60	°C
Humidity Range Non condensing, relative humidity				95	%

Notes:

^[1] The minimum input voltage is software defined with a user selectable range of 3.88V to 5.08V in 80mV increments. Out of the box, the minimum input voltage is set to 4.36V in order for the LiPo battery to be able to properly charge.

^[2] 3G operation

^[3] 2G operation

POWER CONSUMPTION

Parameter	Symbol	Min	Typ	Peak	Unit
Operating Current (uC on, peripherals and radio disabled)	I_{idle}	29.5	33.3	34.6	mA
Operating Current (uC on, cellular on but not connected)	I_{cell_idle}	26.8	42.6	786	mA
Operating Current (uC on, cellular connecting to tower)	$I_{cell_conn_twr}$	59.7	103	592	mA
Operating Current (uC on, cellular connecting to cloud)	$I_{cell_conn_cloud}$	54.0	106	897	mA
Operating Current (uC on, cellular connected but idle)	$I_{cell_cloud_idle}$	30.8	37.0	114	mA
Operating Current (uC on, cellular connected and transmitting)	$I_{cell_cloud_tx}$	49.9	114	914	mA
STOP mode sleep, GPIO wake-up	I_{stop_gpio}	1.72	2.40	3.14	mA
STOP mode sleep, analog wake-up	I_{stop_analog}	5.48	6.03	6.69	mA
STOP mode sleep, RTC wake-up	I_{stop_intrtc}	1.92	2.53	3.12	mA
STOP mode sleep, serial wake-up	I_{stop_usart}	12.5	13.1	13.9	mA
STOP mode sleep, cellular wake-up	I_{stop_cell}	23.8	28.1	93.0	mA
ULP mode sleep, GPIO wake-up	I_{ulp_gpio}	1.91	2.42	2.99	mA
ULP mode sleep, RTC wake-up	I_{ulp_intrtc}	1.94	2.55	3.21	mA
HIBERNATE mode sleep, GPIO wake-up	I_{hib_gpio}	108	114	121	uA
HIBERNATE mode sleep, RTC wake-up	I_{hib_rtc}	108	114	120	uA

¹The min, and particularly peak, values may consist of very short transients. The typical (typ) values are the best indicator of overall power consumption over time. The peak values indicate the absolute minimum capacity of the power supply necessary, not overall consumption.

Current measurements taken at 3.6V via the battery input. For more information about measuring power usage, see [power measurement](#).

RADIO SPECIFICATIONS

The Electron is available in different versions: A 2G version based on u-blox G350 cellular module, two 3G versions based on U260 and U270 modules, and a LTE Cat M1 model (R410M-02B-00 or R410M-02B-03).

Some countries have already stopped supporting 2G, including Australia, Japan, Korea, Singapore, and Taiwan. The cellular carrier used by the Electron no longer supports 2G in New Zealand and Switzerland. The G350 cannot be used in these countries.

The difference between the 3G versions is their operating frequency band which differs based on the country. All of these cellular modules are GSM only and do not support CDMA networks. Both 3G models can fall back to using 2G in areas that support 2G and not 3G.

Note that LTE is LTE Cat M1, not the standard LTE (LTE Cat 1) used by your mobile phone. It is a low-power and low-data-rate variation of LTE for use with IoT devices.

Electron 3G

Module Compatible Countries

U201	Global
U260	United States, Australia, Argentina, Brazil, Canada, Chile, Colombia, Costa Rica, Dominican Republic, El Salvador, Guatemala, Honduras, Mexico, New Zealand, Nicaragua, Panama, Paraguay, Peru, Venezuela
U270	Austria, Bahrain, Belarus, Belgium, Bulgaria, China, Congo, Croatia, Cyprus, Czech Republic, Denmark, Ecuador, Egypt, Estonia, Finland, France, Germany, Ghana, Gibraltar, Greece, Hong Kong, Hungary, Iceland, India, Indonesia, Ireland, Israel, Italy, Japan, Jersey, Kenya, Republic of Korea, Latvia, Lithuania, Luxembourg, Republic of Macedonia, Malaysia, Republic of Moldova, Republic of Montenegro, Netherlands, Nigeria, Norway, Pakistan, Philippines, Poland, Portugal, Qatar, Reunion, Romania, Russian Federation, Rwanda, Saudi Arabia, Republic of Serbia, Seychelles, Sierra Leone, Singapore, Slovakia, Slovenia, South Africa, Spain, Sri Lanka, Swaziland, Sweden, Switzerland, Taiwan, United Republic of Tanzania, Thailand, Turkey, Uganda, Ukraine, United Arab Emirates, United Kingdom, Uruguay, Zambia
R410M	United States, Canada, Mexico

Please be sure to order a board that works in the country where you want to deploy your project.

2G cellular characteristics for G350, U201, U260, and U270 modules:

Parameter	SARA-U201	SARA-U260	SARA-U270	SARA-G350
Protocol stack	3GPP Release 7	GPP Release 7	3GPP Release 7	3GPP Release 99
MS Class	Class B	Class B	Class B	Class B
GSM 850 MHz Band	✓	✓		✓
E-GSM 900 MHz Band	✓		✓	✓
DSC 1800 MHz Band	✓		✓	✓
PCS 1900 MHz Band	✓	✓		✓
Power Class 850/900	Class 4 (33 dBm)	Class 4 (33 dBm)	Class 4 (33 dBm)	Class 4 (33 dBm)
Power Class 1800/1900	Class 1 (30 dBm)	Class 1 (30 dBm)	Class 1 (30 dBm)	Class 1 (30 dBm)

3G cellular characteristics for U201, U260, and U270 modules:

Parameter	SARA-U201	SARA-U260	SARA-U270
Protocol stack	3GPP Release 7	3GPP Release 7	3GPP Release 7
UE Class	Class A	Class A	Class A
Band 5 (850 MHz)	✓	✓	
Band 8 (900 MHz)	✓		✓

Band 1 (2100 MHz)	✓		✓
Band 2 (1900 MHz)	✓	✓	
Power Class	Class 3 (24 dBm)	Class 3 (24 dBm)	Class 3 (24 dBm)

u-blox SARA-R410M-02B-00 or R410M-02B-03

Parameter	Value
Protocol stack	3GPP Release 13
RAT	LTE Cat M1 Half-Duplex
LTE FDD Bands	Band 12 (700 MHz)
	Band 28 (700 MHz)
	Band 13 (750 MHz)
	Band 20 (800 MHz)
	Band 5 (850 MHz)
	Band 8 (900 MHz)
	Band 4 (1700 MHz)
	Band 3 (1800 MHz)
	Band 2 (1900 MHz)
Power class	Class 3 (23 dBm)

I/O CHARACTERISTICS

These specifications are based on the STM32F205RGT6 datasheet, with reference to Electron pin nomenclature.

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Standard I/O input low level voltage	V_{IL}		-0.3		$0.28 \cdot (V_{3V3} - 2) + 0.8$	V
I/O FT ^[1] input low level voltage	V_{IL}		-0.3		$0.32 \cdot (V_{3V3} - 2) + 0.75$	V
Standard I/O input high level voltage	V_{IH}		$0.41 \cdot (V_{3V3} - 2) + 1.3$		$V_{3V3} + 0.3$	V
I/O FT ^[1] input high level voltage	V_{IH}	$V_{3V3} > 2V$	$0.42 \cdot (V_{3V3} - 2) + 1$		5.5	V
	V_{IH}	$V_{3V3} \leq 2V$	$0.42 \cdot (V_{3V3} - 2) + 1$		5.2	V
Standard I/O Schmitt trigger voltage hysteresis ^[2]	V_{hys}		200			mV
I/O FT Schmitt trigger voltage hysteresis ^[2]	V_{hys}		$5\% V_{3V3}$ ^[3]			mV
Input leakage current ^[4]	I_{lkg}	$GND \leq V_{io} \leq V_{3V3}$ GPIOs			± 1	μA
Input leakage current ^[4]	I_{lkg}	R_{PU}		$V_{io} = 5V$, I/O FT	3	μA
Weak pull-up equivalent resistor ^[5]	R_{PU}	$V_{io} = GND$	30	40	50	k Ω
Weak pull-down equivalent resistor ^[5]	R_{PD}	$V_{io} = V_{3V3}$	30	40	50	k Ω
I/O pin capacitance	C_{IO}			5		pF
DAC output voltage (buffers enabled by default)	V_{DAC}		0.2		$V_{3V3} - 0.2$	V
DAC output resistive load (buffers enabled by default)	R_{DAC}		5			k Ω
DAC output capacitive load (buffers enabled by default)	C_{DAC}				50	pF

Notes:

^[1] FT = Five-volt tolerant. In order to sustain a voltage higher than $V_{3V3} + 0.3$ the internal pull-up/pull-down resistors must be disabled.

^[2] Hysteresis voltage between Schmitt trigger switching levels. Based on characterization, not tested in production.

^[3] With a minimum of 100mV.

^[4] Leakage could be higher than max. if negative current is injected on adjacent pins.

^[5] Pull-up and pull-down resistors are designed with a true resistance in series with switchable PMOS/NMOS. This PMOS/NMOS contribution to the series resistance is minimum (~10% order).

Mechanical specifications

DIMENSIONS AND WEIGHT

- Width = 0.8"
- Height = 0.65"
- Length = 2.05"
- Weight = 10 grams

MATING CONNECTORS

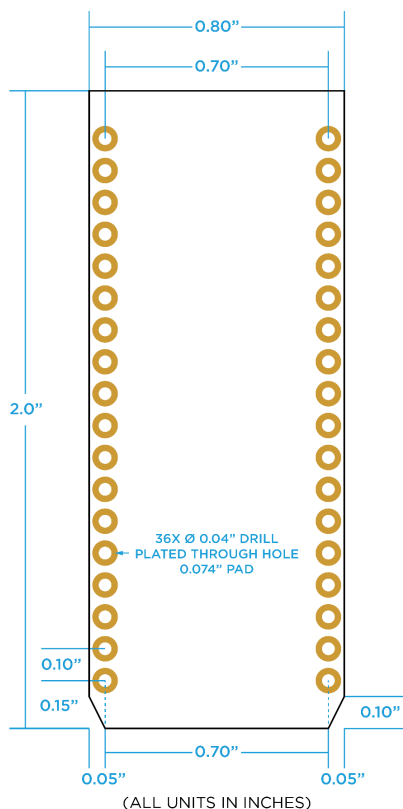
The Electron can be mounted with (qty 2) 18-pin single row 0.1" female headers. Typically these are 0.335" (8.5mm) tall, but you may pick a taller one if desired. When you search for parts like these it can be difficult to navigate the thousands of parts available online so here are a few good choices for the Electron:

| Description | MFG | MFG Part Number | Distributor | |:-|:-|:-| | 18-pin 0.1" (2.54mm) Female Header (Tin) | Sullins Connector Solutions | [PPTC181LFBN-RC](#) | DigiKey | | 18-pin 0.1" (2.54mm) Female Header (Tin) | 3M | [929974-01-18-RK](#) | DigiKey | | 18-pin 0.1" (2.54mm) Female Header (Tin) | Harwin | [M20-7821846](#) | Mouser |

You may also use other types, such as reverse mounted (bottom side SMT) female headers, low profile types, etc..

RECOMMENDED PCB LAND PATTERN

The Electron can be mounted with 0.1" 18-pin female header receptacles using the following PCB land pattern:



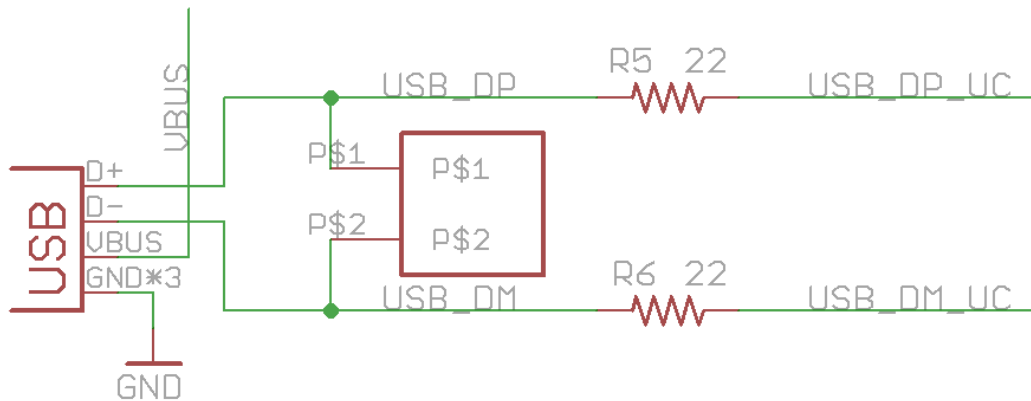
An Electron part for EAGLE can be found in the [Particle EAGLE library](#)

Schematic

All of the Electron hardware design files are open source and available under a Creative Commons Public License. The schematic and PCB designs were made using EAGLE CAD. You can access these files [here](#).

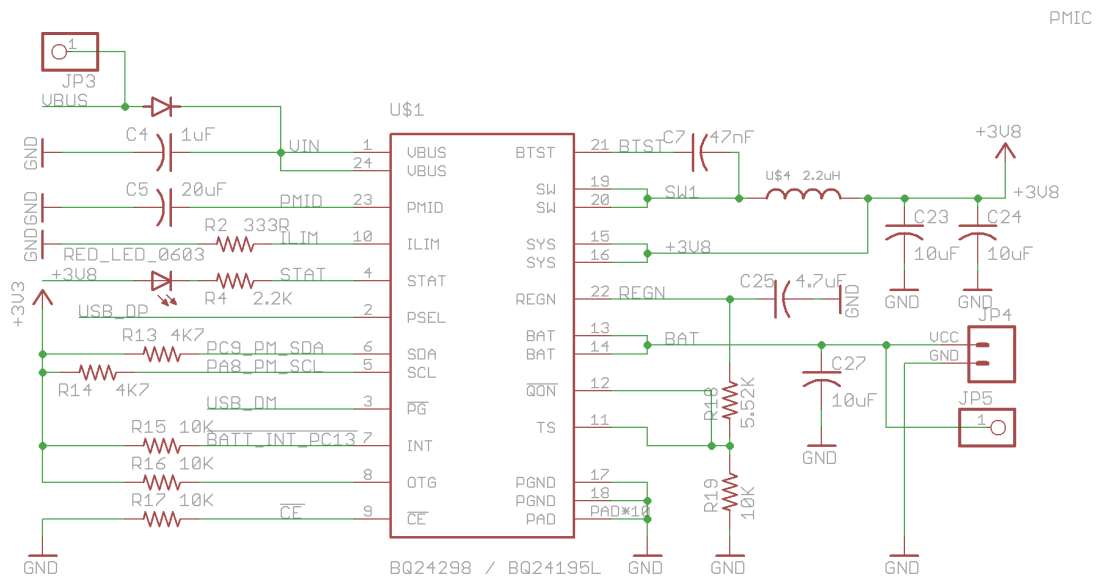
Note: Clone or Download the complete repository as a ZIP file to avoid corrupted data in Eagle files.

USB



The USB data lines are terminated with 22 Ohm resistors. These data pins are also exposed via small through holes next to the USB connector and are labeled D+ and D-. The VBUS (+5VDC VCC of the USB port) is fed to the PMIC via a 3Amp Schottky diode ([SS3P3](#)). The VBUS pin is also available via the unpopulated header hole on the top-right side of the Electron.

PMIC (POWER MANAGEMENT INTEGRATED CIRCUIT)



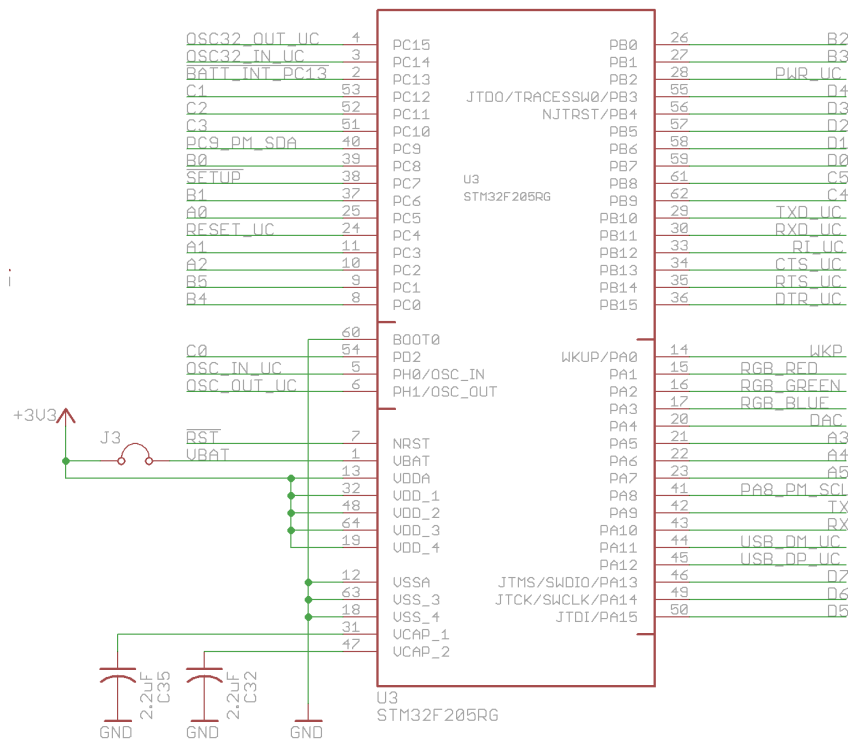
The Electron uses TI's [BQ24195](#) as the power management and charging unit. This PMIC intelligently sources power from either the VIN pin, the USB port and/or the LiPo battery. When all

the power sources as connected, the unit tries to source power from the USB or VIN as default and continues to charge the LiPo battery. When the battery is completely charged, the power is then sourced from USB/VIN alone. If there is a power deficit (which generally occurs during cellular radio transmission), the additional power is then sourced from the battery as required. The unit can also seamlessly switch back to the battery when other sources of power are suddenly removed.

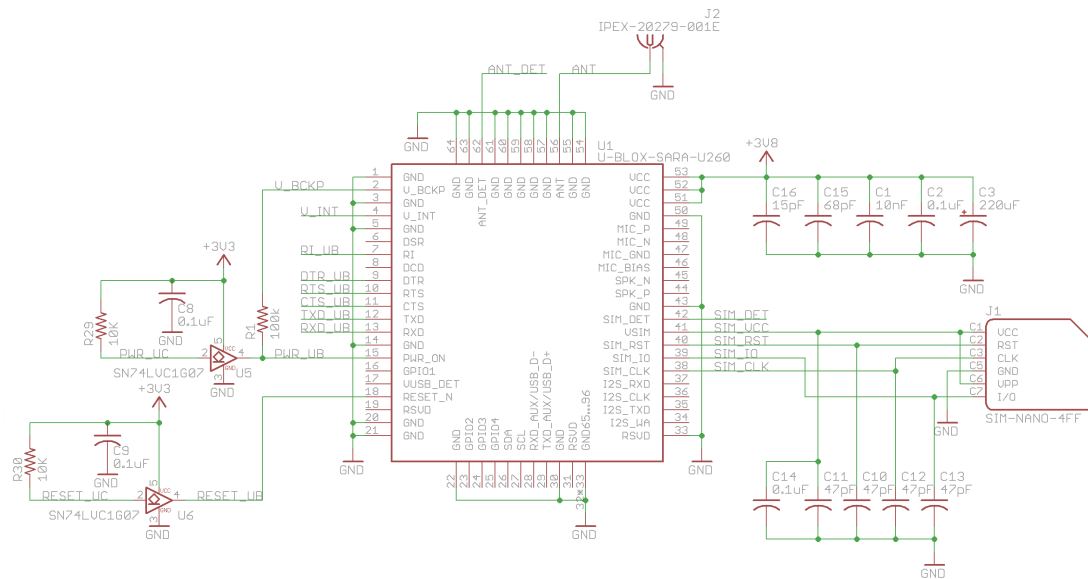
The DP data pin of the USB is used by the PMIC to detect the presence of a USB power source. It then adjusts the charge current and the limit based on the type of USB power source it detects. This does not always happen successfully since there are a lot of USB hubs and chargers out there that do not meet the USB design guidelines. If the detection is unsuccessful, the PMIC defaults to a 500mA current limit. A user can always adjust these parameters via software.

The microcontroller communicates with the PMIC via an I2C interface (pins PC9 and PA8). This interface allows the microcontroller to read the status of the PMIC and set its various parameters.

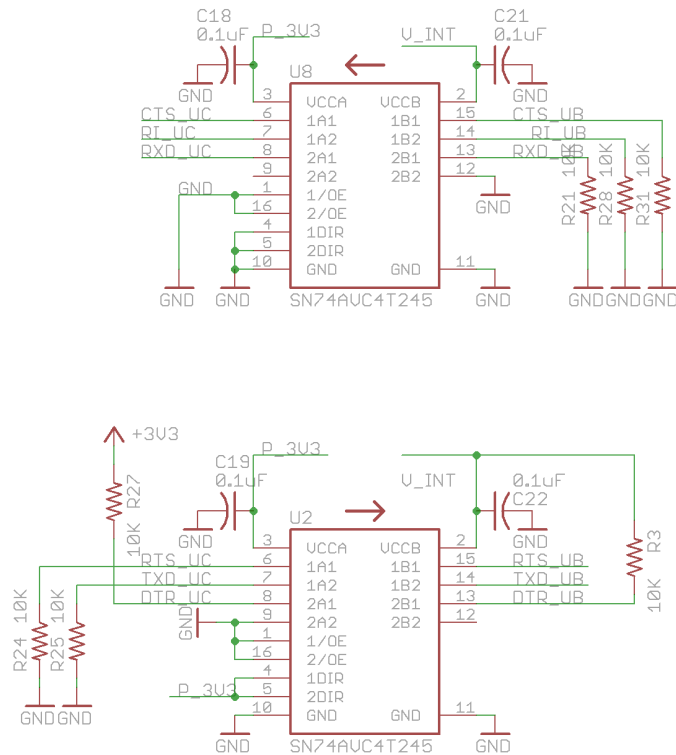
MICROCONTROLLER



The Electron uses ST Microelectronics's STM32F205RGT6 ARM Cortex M3 microcontroller running at 120MHz.

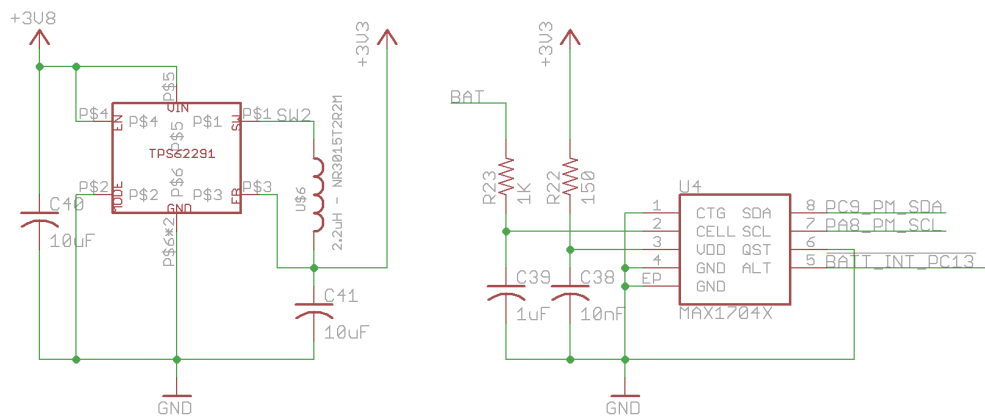


The u-blox cellular module talks to the microcontroller over a full-duplex USART interface using a standard set of AT commands. The SIM (Nano 4FF) card is directly connected to the u-blox. The power to the SIM card is also provided by the cellular module.



Since u-blox module's communication interface operates at 1.8VDC, while the STM32F205RGT6 microcontroller operates at 3.3VDC, we need voltage translators in-between them. This is achieved with two [SN74AUC4T245](#) non-inverting buffers. The default state of the USART pins is set with the help of pull-up and pull-down resistors, and the unused input pins are tied to GND.

3.3V REGULATOR AND FUEL GAUGE

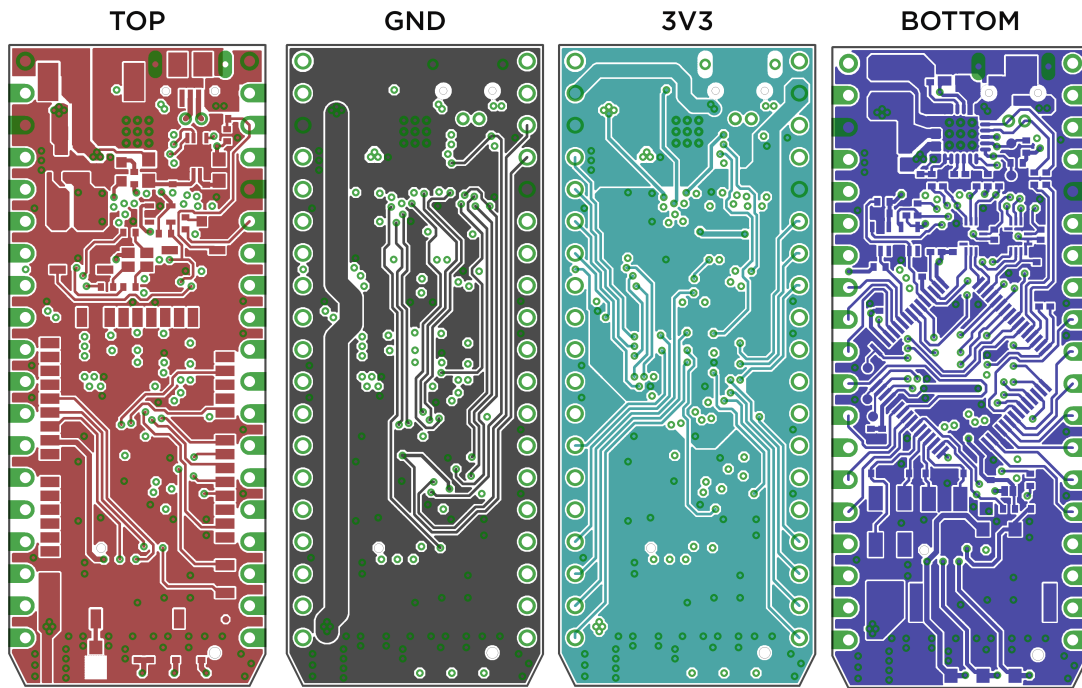


The output (3.8V net) of the PMIC is fed directly to the u-blox cellular module and a 3.3VDC high efficiency switching regulator ([TPS62290](#)). This 3.3VDC regulator helps power the microcontroller, fuel gauge and the buffers.

The Electron employs a [MAX17043](#) fuel gauge to monitor the LiPo battery voltage and it's state of charge. The microcontroller communicates with it over an I2C interface (same channel as the PMIC).

Layout

The Electron uses a four layer circuit board. Top layer consists of a signal layer followed by ground (GND), 3.3V power (3V3), and bottom signal.



Bill of materials

QTY	Device	Value	Package	Designator	Manufacturer	MFG. Part #	
							14 CAPACITOR
		0.1uF, 6.3V, 10%	0402	C14, C17, C18, C19, C2, C20, C21, C22, C29, C30, C31, C6, C8, C9	Fenghua		
		0402B104K160NT	2	CAPACITOR	10nF, 6.3V, 10%	0402 C1,C38	Fenghua 0402B103K500NT 5
		CAPACITOR	10uF, 6.3V, 10%	0603	C23, C24, C27, C40, C41	Yageo CC0603KRX5R5BB106	2
		CAPACITOR	12pF, 6.3V, 10%	0402	C33, C34	Fenghua 0402CG120J500NT	1 CAPACITOR 15pF,
		6.3V, 10%	0402	C16	Fenghua 0402CG150J500NT	2 CAPACITOR 1uF, 6.3V, 10%	0402 C37, C39
		Fenghua 0402X105K6R3NT	1 CAPACITOR 1uF, 25V, 10%	0603	C4	Yageo	
		CC0603KRX5R8BB105	2 CAPACITOR 2.2uF, 6.3V, 10%	0402	C32, C35	Yageo	
		CC0402KRX5R5BB225	2 CAPACITOR 20pF, 6.3V, 10%	0402	C26, C28	Fenghua	
		0402CG200J500NT	1 CAPACITOR 22uF, 6.3V, 10%	0603	C5	Samsung CL10A226KQ8NRNE	1
		CAPACITOR	220uF, 6.3V, 10%	2312 (6032 metric)	C3	AVX TAJC227K006	1 CAPACITOR 4.7uF,
		6.3V, 10%	0402	C36	Samsung CL05A475KQ5NRNC	1 CAPACITOR 4.7uF, 6.3V, 10%	0603 C25
		Yageo CC0603KRX5R5BB475	1 CAPACITOR 47nF, 6.3V, 10%	0402	C7	Fenghua	
		0402B473K160NT	4 CAPACITOR 47pF, 6.3V, 10%	0402	C10, C11, C12, C13	Fenghua	
		0402CG470J500NT	1 CAPACITOR 68pF, 6.3V, 10%	0402	C15	Fenghua 0402CG680J500NT	1
		CONNECTOR	2-pin SMD, 2-pin, Vertical	JP4	Kaweei CW2001-02T-M01-D	1 CONNECTOR 1x18	
		1x18, 0.1" pitch	JP1	Kaweei CP25411-18G-S116A-A	1 CONNECTOR 1x18 1x18, 0.1" pitch	JP2	
		Kaweei CP25411-18G-S116B-A	1 CONNECTOR		USB-MICROB-SLOT-HOLE	X1	Kaweei CMCUSB-
		5BFM2G-01-D	1 CONNECTOR		SMD	J2	Kaweei P1163-0140R
		1 CONNECTOR			10mm x		
		12.3mm	J1	Kaweei CSIM2545-06S-D	1 CRYSTAL 26MHz, <±20ppm	4-SMD, 5.0 x 3.2mm	Y2
		Song Ji SJSMD5026M00018F20	1 CRYSTAL 32.768KHz, <±20ppm	2-SMD, 1.5 x 3.2mm	Y1	Song	Ji SJ FC1332K012F520P
		1 DIODE	30V, 3A	DO-220AA	U\$3	Vishay SS3P3-M3/84A	2 IC - Buffer
		SC-70-5	U5, U6	Texas Instruments SN74LVC1G07DCKR	1 IC - Fuel Gauge	TDFN-8	U4
		Maxim MAX17043G+T	1 IC - Cell Module	16 x 26 x 3mm	U1	u-blox SARA-G350SARA-	
		U260SARA-U270	2 IC - Buffer	16-UQFN	U2, U8	Texas Instruments SN74AVC4T245RSVR	1
		IC - Microcontroller	LQFP64	U3	ST Microelectronics STM32F205RGT6	1 IC - PMIC	24VQFN
		U\$1	Texas Instruments BQ24195RGER	1 IC - MOSFET	SC70-5	U9	Texas Instruments
		TPS22942DCKR	1 IC - 3V3 Reg	1A	6-SON (2x2)	U\$5	Texas Instruments TPS62291DRVR
		1 INDUCTOR	2.2uH, 1.5A, 20%	3.0 mm x 3.0 mm	U\$6	Taiyo Yuden NR3015T2R2M	1 INDUCTOR
		2.2uH, 4A	4.45mm x 4.06mm	U\$4	Bourns, Inc. SRP4020TA-2R2M	1 LED Blue	0603 LED1
		Everlight 19-217/BHC-ZL1M2RY/3T	1 LED Red	0603	LED3	Everlight 19-217/R6C-AL1M2VY/3T	1
		LED RGB	4-PLCC (3.2 mm x 2.8mm)	LED2	Cree CLMVB-FKA-CFHEHLCBB7A363	1	
		RESISTOR	0R, 1/16W	0201	R32	Fenghua	1 RESISTOR 100k, 1/16W, 5%
		0402	R1	Fenghua RC-			
		02W104JT	14 RESISTOR 10K, 1/16W, 5%	0402	R3, R8, R15, R16, R17, R19, R21, R24, R25, R27, R28,		
		R29, R30, R31	Fenghua RC-02W103JT	1 RESISTOR 150R, 1/4W, 1%	0603	R22	Vishay
		CRCW0603150RFKEAHP	1 RESISTOR 1K, 1/16W, 5%	0603	R23	Fenghua RC-03K102JT	4
		RESISTOR 1K, 1/16W, 5%	0402	R10, R11, R12, R9	Fenghua RC-02W102JT	1 RESISTOR 2.2K, 1/16W,	
		5%	0402	R4	Fenghua RC-02W222JT	2 RESISTOR 22R, 1/16W, 1%	0402 R5, R6
		Fenghua RC-					
		02W22R0FT	1 RESISTOR 330R, 1/16W, 1%	0402	R2	Fenghua RC-02W3300FT	2 RESISTOR
		4K7, 1/16W, 5%	0402	R13, R14	Fenghua RC-02W472JT	1 RESISTOR 5.49K, 1/16W, 1%	0402 R18
		Fenghua RC-02W5491FT	2 SWITCH 160gF	3.6mm x 3.1mm	MODE, RESET	Haoyu TS-1185A-C	

Country compatibility

Country	Model	Technologies	Carriers
Afghanistan	ELC314	2G, 3G	MTN
Albania	ELC314	2G, 3G	Eagle, Telekom, Vodafone
Algeria	ELC314	2G, 3G	Mobilis, Ooredoo
Anguilla	ELC314	2G, 3G	Flow
Antigua and Barbuda	ELC314	2G, 3G	Flow
Argentina	ELC314	2G, 3G	Claro, Movistar, Personal
Armenia	ELC314	2G, 3G	Beeline, Ucom
Aruba	ELC314	2G, 3G	Setar
Austria	ELC314	2G	3 (Drei), A1, T-Mobile
Azerbaijan	ELC314	2G, 3G	Azercell, Bakcell, NAR Mobile
Bahamas	ELC314	2G, 3G	Aliv, BTC Bahamas
Bahrain	ELC314	2G	Zain
Bangladesh	ELC314	2G, 3G	Bangalink, GrameenPhone
Barbados	ELC314	2G, 3G	Flow
Belarus	ELC314	2G, 3G	A1
Belgium	ELC314	2G	Base, Orange, Proximus
Belize	ELC314	3G	Smart
Bolivia	ELC314	2G, 3G	Viva
Bosnia and Herzegovina	ELC314	2G, 3G	HT Eronet
Brunei	ELC314	3G	DST
Bulgaria	ELC314	2G, 3G	A1, Telenor, Vivacom
Burkina Faso	ELC314	2G, 3G	Orange
Cambodia	ELC314	2G, 3G	Metfone
Canada	ELC404	M1	Bell Mobility, Rogers Wireless, Telus
Cayman Islands	ELC314	2G, 3G	Flow
Chad	ELC314	2G, 3G	Airtel
Chile	ELC314	2G, 3G	Claro, Entel, Movistar
Colombia	ELC314	2G, 3G	Movistar, Tigo
Congo (Brazzaville)	ELC314	2G, 3G	Airtel
Congo (Kinshasa)	ELC314	2G, 3G	Airtel
Costa Rica	ELC314	3G	Movistar
Côte d'Ivoire	ELC314	2G, 3G	MTN
Croatia	ELC314	2G, 3G	Hrvatski Telekom, Tele2
Cyprus	ELC314	2G, 3G	MTN
Czechia	ELC314	2G	T-Mobile, Vodafone
Denmark	ELC314	2G	TDC, Telenor, Telia
Dominica	ELC314	2G, 3G	Flow
Dominican Republic	ELC314	2G, 3G	Altice Dominicana, Viva
Ecuador	ELC314	2G, 3G	Claro, Movistar
Egypt	ELC314	2G, 3G	Etisalat, Orange
El Salvador	ELC314	2G, 3G	Claro, Telefonica
Estonia	ELC314	2G, 3G	Elisa, Tele2, Telia

eSwatini	ELC314	2G, 3G	MTN
Ethiopia	ELC314	2G, 3G	Ethio Telecom
Faroe Islands	ELC314	2G, 3G	Faroese Telecom, Vodafone
Finland	ELC314	2G	Elisa
France	ELC314	2G, 3G	Bouygues, Free Mobile, Orange, SFR
French Guiana	ELC314	2G, 3G	Digicel
Gabon	ELC314	2G, 3G	Airtel
Georgia	ELC314	2G, 3G	Beeline, Geocell
Germany	ELC314	2G	O2, Telekom
Ghana	ELC314	2G, 3G	AirtelTigo, MTN, Vodafone
Gibraltar	ELC314	2G, 3G	Gibtel
Greece	ELC314	2G	Cosmote, Wind
Grenada	ELC314	2G	Flow
Guadeloupe	ELC314	2G, 3G	Orange
Guatemala	ELC314	2G, 3G	Claro, Movistar
Guinea	ELC314	2G, 3G	MTN
Guinea-Bissau	ELC314	2G, 3G	MTN
Guyana	ELC314	2G	Digicel
Haiti	ELC314	2G, 3G	Digicel
Honduras	ELC314	2G, 3G	Claro, Tigo
Hong Kong	ELC314	3G	CSL, SmarTone
Hungary	ELC314	2G, 3G	Telenor
Indonesia	ELC314	2G, 3G	Indosat, Telkomsel, XL Axiata
Ireland	ELC314	2G, 3G	Meteor, O2, Vodafone
Israel	ELC314	2G, 3G	Hot Mobile
Italy	ELC314	2G	TIM, Vodafone, Wind
Jordan	ELC314	2G, 3G	Zain
Kazakhstan	ELC314	2G, 3G	Beeline, K-Cell
Kenya	ELC314	2G, 3G	Airtel
Kuwait	ELC314	2G, 3G	Viva, Zain
Kyrgyzstan	ELC314	2G	Beeline
Latvia	ELC314	2G	Bite, LMT, Tele2
Liechtenstein	ELC314	2G, 3G	Mobilkom, Orange
Lithuania	ELC314	2G	Bite, Omnitel, Tele2
Luxembourg	ELC314	2G	Orange, POST, Tango
Malawi	ELC314	2G, 3G	Airtel
Malaysia	ELC314	2G, 3G	Celcom, DiGi, Maxis
Malta	ELC314	2G, 3G	Go Mobile, Vodafone
Mexico	ELC404	M1	AT&T, Telcel
Moldova	ELC314	2G, 3G	Moldcell, Orange
Mongolia	ELC314	2G, 3G	Mobicom, Unitel
Montenegro	ELC314	2G, 3G	T-Mobile, Telenor
Mozambique	ELC314	2G, 3G	Vodacom
Myanmar	ELC314	2G, 3G	MPT
Namibia	ELC314	2G, 3G	Telecom Namibia

Netherlands	ELC314	2G, 3G	T-Mobile, Vodafone
Nicaragua	ELC314	2G, 3G	Movistar
Nigeria	ELC314	2G, 3G	9mobile, Airtel, Glo, MTN
Norway	ELC314	2G, 3G	TDC
Pakistan	ELC314	2G, 3G	Mobilink, Ufone, Warid
Palestine	ELC314	2G, 3G	Jawwal
Panama	ELC314	2G, 3G	Digicel, Movistar
Papua New Guinea	ELC314	2G, 3G	bmoblie
Paraguay	ELC314	2G, 3G	Claro, Personal, Tigo, Vox
Peru	ELC314	2G, 3G	Claro, Entel, Movistar
Philippines	ELC314	2G, 3G	Globe, Smart
Poland	ELC314	2G, 3G	Orange, Play, Plus, T-Mobile
Portugal	ELC314	2G, 3G	NOS, TMN, Vodafone
Qatar	ELC314	2G, 3G	Ooredoo, Vodafone
Romania	ELC314	2G	Orange, Telekom Romania, Vodafone
Rwanda	ELC314	2G, 3G	Airtel, MTN
Saint Kitts and Nevis	ELC314	2G, 3G	Flow
Saint Lucia	ELC314	2G, 3G	Flow
Saint Vincent and the Grenadines	ELC314	3G	Flow
Serbia	ELC314	2G, 3G	Telenor, VIP
Sint Maarten	ELC314	2G, 3G	TelCell
Slovakia	ELC314	2G	O2, Orange, Telekom
Slovenia	ELC314	2G, 3G	A1, Mobitel
South Africa	ELC314	2G, 3G	Cell C, MTN, Vodacom
South Korea	ELC314	3G	KT, SK Telecom
South Sudan	ELC314	2G, 3G	MTN
Spain	ELC314	2G, 3G	Orange, Telefonica, Vodafone, Yoigo
Sri Lanka	ELC314	2G, 3G	Dialog, Mobitel
Suriname	ELC314	2G, 3G	Telesur
Sweden	ELC314	2G	3 (Tre), Telia
Switzerland	ELC314	3G	Salt
Taiwan	ELC314	3G	T Star
Tajikistan	ELC314	2G, 3G	Beeline, Tcell
Tanzania	ELC314	2G, 3G	Airtel
Thailand	ELC314	2G, 3G	AIS, DTAC
Trinidad and Tobago	ELC314	2G, 3G	Digicel, TSTT
Tunisia	ELC314	2G, 3G	Orange Tunisie, Tunisie Telecom
Turks and Caicos Islands	ELC314	2G, 3G	Flow
Uganda	ELC314	2G, 3G	Africell, Airtel, MTN
Ukraine	ELC314	2G, 3G	Kyivstar, Life, MTS
United Kingdom	ELC314	2G	EE
United States	ELC404	M1	AT&T, T-Mobile (USA)
Uruguay	ELC314	2G, 3G	Antel, Movistar
Uzbekistan	ELC314	2G, 3G	Beeline
Venezuela	ELC314	3G	Movistar

Vietnam	ELC314	3G	MobiFone, Vinaphone
Virgin Islands (British)	ELC314	2G, 3G	CCT, Flow
Zambia	ELC314	2G, 3G	Airtel

Ordering information

Electrons are available from store.particle.io in single quantities in 2G, and 3G versions.

SKU	Description	Region	Modem	EtherSIM	Lifecycle	Replacement
E270TRAY50	Electron 2G/3G (EMEA), Tray [x50]	EMEAA	U270		NRND	B524MTY
ELC314TY	Electron 2G/3G (Global - U201) , Tray [x50]	Global	U201	✓	NRND	
ASSET2GV2	Asset Tracker 2G	Global	G350		Deprecated	
ASSET3G260V2	Asset Tracker 3G (Americas/Aus)	Americas	U260		Deprecated	
ASSET3G270V2	Asset Tracker 3G (Eur/Asia/Afr)	EMEAA	U270		Deprecated	
E260KIT	Electron 2G/3G (Americas/Aus) Starter Kit, [x1]	Americas	U260		Deprecated	BRN404XKIT
E260TRAY50	Electron 2G/3G (Americas/Aus), Tray [x50]	Americas	U260		Deprecated	BRN404XTRAY50
E270KIT	Electron 2G/3G (EMEA) Starter Kit, [x1]	EMEAA	U270		Deprecated	B524MEA
E350KIT	Electron 2G Kit (Global)	Global	G350		Deprecated	B524MEA
E350TRAY50	Electron 2G (Global), Tray [x50]	Global	G350		Deprecated	B524MTY
ELC402EA	Electron LTE CAT-M1 (NorAm), [x1]	NORAM	R410		Deprecated	BRN404XKIT
ELC402TY	Electron LTE CAT-M1 (NorAm), Tray [x50]	NORAM	R410		Deprecated	BRN404XTRAY50
ELC404TY	Electron LTE CAT-M1 (NorAm, EtherSIM), Tray [x50]	NORAM	R410	✓	Deprecated	
SNSRKIT3G260	Electron 3G (Americas/Aus) Sensor Kit, [x1]	Americas	U260		Deprecated	
SNSRKIT3G270	Electron 3G (Eur/Asia/Afr) Sensor Kit, [x1]	EMEAA	U270		Deprecated	

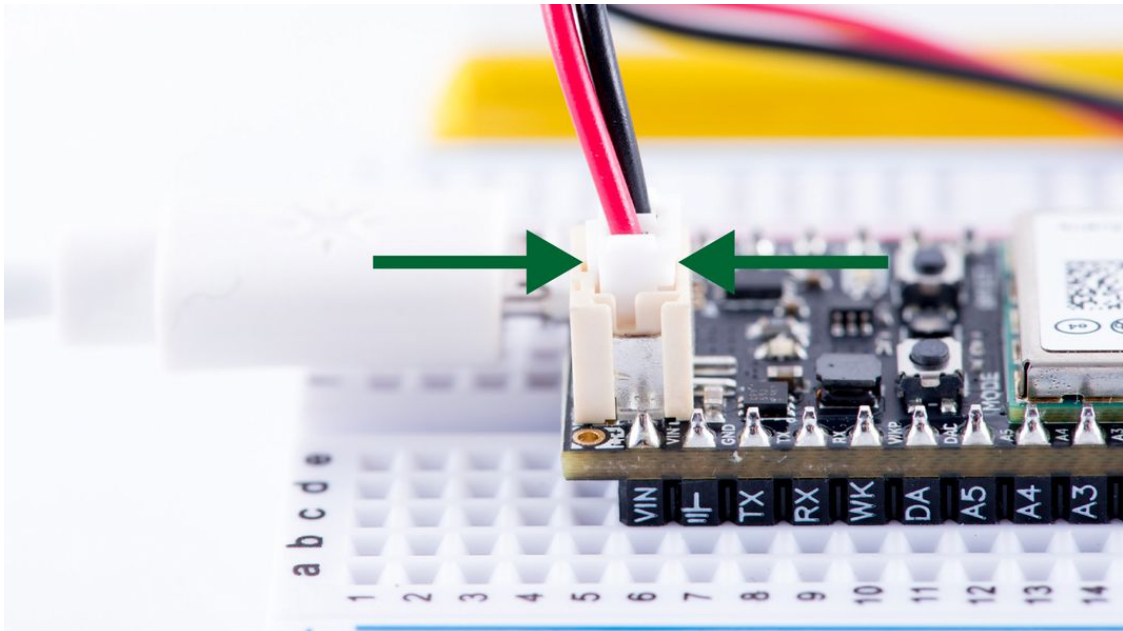
Product handling

ESD PRECAUTIONS

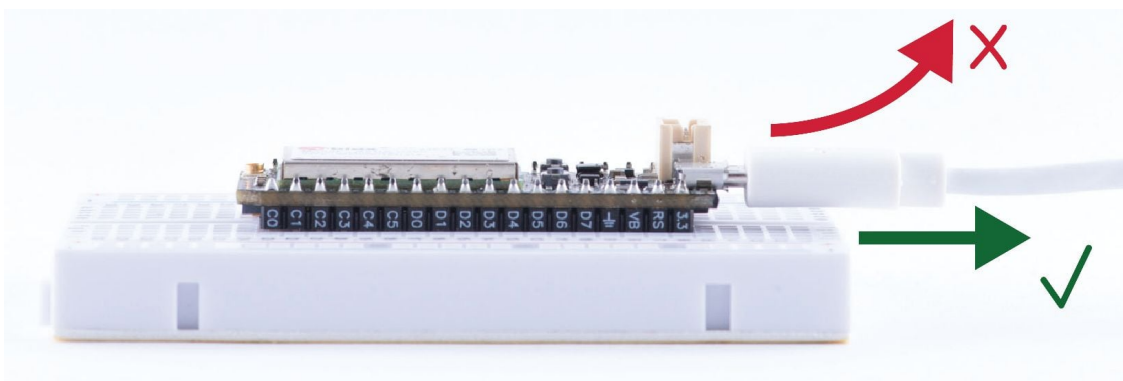
The Electron contains highly sensitive electronic circuitry and is an Electrostatic Sensitive Device (ESD). Handling a Electron without proper ESD protection may destroy or damage it permanently. Proper ESD handling and packaging procedures must be applied throughout the processing, handling and operation of any application that incorporates Electrons. ESD precautions should be implemented on the application board where the Electron is mounted. Failure to observe these precautions can result in severe damage to the Electron!

CONNECTORS

There are three connectors on the Electron that will get damaged with improper usage. The JST connector on the circuit board, where you plug in the LiPo battery, is very durable but the connector on the battery itself is not. When unplugging the battery, take extra precaution to **NOT** pull the connector using the wires, but instead hold the plug at its base to avoid putting stress on the wires. This can be tricky with bare hands - nose pliers are your friend here.



The micro B USB connector on the electron is soldered on the PCB with large surface pads as well as couple of through hole anchor points. Despite this reinforcement, it is very easy to rip out the connector if too much stress is put on in the vertical direction.



The U.FL antenna connector is not designed to be constantly plugged and unplugged. The antenna pin is static sensitive and you can destroy the radio with improper handling. A tiny dab of

glue (epoxy, rubber cement, liquid tape or hot glue) on the connector can be used securely hold the plug in place.

BREADBOARDING

The breadboard provided with the Electron is specifically designed to require low insertion force. This makes it easy to plug the Electron in and out of the breadboard. If you end up using a different breadboard, remember that it may require more force. In this case, always remember to pinch-hold your precious Electron by the sides (along the header pins) when plugging-unplugging and **not** by the USB connector (don't be this person).

Default settings

The Electron comes pre-programmed with a bootloader and a user application called Tinker. This application works with an iOS and Android app also named Tinker that allows you to very easily toggle digital pins, take analog and digital readings and drive variable PWM outputs.

The bootloader allows you to easily update the user application via several different methods, USB, OTA, Serial Y-Modem, and also internally via the Factory Reset procedure. All of these methods have multiple tools associated with them as well.

You may use the [Particle Web IDE](#) to code, compile and flash a user application OTA (Over The Air). [Particle Workbench](#) is a full-featured desktop IDE for Windows, Mac, and Linux based on VSCode and supports both cloud-based and local gcc-arm compiles. The [Particle CLI](#) provides a command-line interface for cloud-based compiles and flashing code over USB.

Glossary

Term	Definition
SMPS	Switch Mode Power Supply
SIM	Subscriber Identity Module (Size: Nano 4FF)
RF	Radio Frequency
SMT	Surface Mount Technology (often associated with SMD which is a surface mount device).
LED	Light Emitting Diode
RGB LED	Red green and blue LEDs combined and diffused in one package.
USB	Universal Serial Bus
Quiescent current	Current consumed in the deepest sleep state.
FT	Five-tolerant; Refers to a pin being tolerant to 5V.
3V3	+3.3Volt; The regulated +3.3V supply rail. Also used to note a pin is only 3.3V tolerant.
PMIC	Power Management Integrated Circuit
LiPo	Lithium-ion Polymer Battery
GSM	Global System for Mobile Communications
CDMA	Code Division Multiple Access
OTA	Over The Air; describing how firmware is transferred to the device.
uC	Microcontroller

FCC ISED CE warnings and end product labeling requirements

Federal Communication Commission Interference Statement This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

FCC Caution: Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate this equipment. This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference, and
2. This device must accept any interference received, including interference that may cause undesired operation.

FCC Radiation Exposure Statement: This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. This transmitter module must not be co-located or operating in conjunction with any other antenna or transmitter. This End equipment should be installed and operated with a minimum distance of 20 centimeters between the radiator and your body.

IMPORTANT NOTE: In the event that these conditions can not be met (for example certain laptop configurations or co-location with another transmitter), then the FCC authorization is no longer considered valid and the FCC ID can not be used on the final product. In these circumstances, the OEM integrator will be responsible for re-evaluating the end product (including the transmitter) and obtaining a separate FCC authorization.

End Product Labeling The final end product must be labeled in a visible area with the following:

Contains FCC ID:

- XPYSARAG350 (For 2G Electron using the G350 module)
- XPYSARAU201 (For 3G Electron using the U201 module)
- XPYSARAU260 (For 3G Electron using the U260 module)
- XPYSARAU270 (For 3G Electron using the U270 module)
- XPY2AGQN4NNN (For LTE Electron module using the R410 module)
- *Manual Information to the End User** The OEM integrator has to be aware not to provide information to the end user regarding how to install or remove this RF module in the user's manual of the end product which integrates this module.

Canada Statement This device complies with Industry Canada's licence-exempt RSSs. Operation is subject to the following two conditions:

1. This device may not cause interference; and
2. This device must accept any interference, including interference that may cause undesired operation of the device.

Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence.

L'exploitation est autorisée aux deux conditions suivantes:

1. l'appareil ne doit pas produire de brouillage;
2. l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

Caution Exposure: This device meets the exemption from the routine evaluation limits in section 2.5 of RSS102 and users can obtain Canadian information on RF exposure and compliance. Le dispositif répond à l'exemption des limites d'évaluation de routine dans la section 2.5 de RSS102 et les utilisateurs peuvent obtenir des renseignements canadiens sur l'exposition aux RF et le respect.

The final end product must be labelled in a visible area with the following: The Industry Canada certification label of a module shall be clearly visible at all times when installed in the host device, otherwise the host device must be labelled to display the Industry Canada certification number of the module, preceded by the words "Contains transmitter module", or the word "Contains", or similar wording expressing the same meaning, as follows:

Contains transmitter module ISED:

- 8595A-SARAG350 (For 2G Electron using the G350 module)
- 8595A-SARAU260 (For 3G Electron using the U260 module)
- 8595A-SARAU270 (For 3G Electron using the U270 module)
- 8595A-2AGQN4NNN (For LTE Electron module using the R410 module)

This End equipment should be installed and operated with a minimum distance of 20 centimeters between the radiator and your body. Cet équipement devrait être installé et actionné avec une distance minimum de 20 centimètres entre le radiateur et votre corps.

The end user manual shall include all required regulatory information/warning as shown in this manual.

For an in-depth review on certifications, please click [here](#).

Revision history

Revision	Date	Author	Comments
v001	20-Jan-2016	MB	Initial release
v002	24-March-2016	MB	Added: Memory map, DAC limits, SIM card size, SWD pin locations. Updated: Power section, pin diagram, block diagram, operating conditions.
v003	12-Sept-2016	BW	Error in Cellular off operating current, changed from 2-15mA to 47-50mA. Also qualified these current readings with uC on/off. Updated the Pin Description section. Updated Mating connectors section.
v004	27-Oct-2016	BW	Replaced one STM32F205RGY6 with STM32F205RGT6, and replaced all STM32 mentions with full part number STM32F205RGT6
v005	14-Aug-2017	BW	Updated DCD layout and Memory Map, renamed SPI1_*/SPI3_* to match Particle API instead of STM32 pin names to avoid confusion (now SPI, SPI1 and SPI2), updated the Pin Description section and added high resolution pinout PDF, updated LED Status section, VBAT info, added Power the Electron without a battery section
v006	31-Jul-2019	RK	Added LTE information
v007	16-Sep-2020	RK	Added power consumption information
v008	24-Feb-2021	RK	Added ELC314 information
v009	10-Sep-2021	RK	Changed wording of peak vs. max current
v010	14-Mar-2022	RK	Added deprecation warning
v011	31-Jan-2023	RK	Added Device OS versions

Known errata

We are tracking [known errata with this datasheet here](#). These issues/errors in the datasheet will be resolved in subsequent revisions.

Contact

Web

<https://www.particle.io>

Community Forums

<https://community.particle.io>